Calculus II

Tutorial Problems #3 03–07 Mar 2025

◇ **Topics:**

- Partial derivatives
- Chain rule
- Implicit differentiation

In Problems 1–2, find f_x, f_y and f_z.

Problem 1.

$$f(x, y, z) = x - \sqrt{y^2 + z^2}.$$

Problem 2.

$$f(x, y, z) = \ln(x + 2y + 3z).$$

Problem 3. Use the limit definition of partial derivative to compute the partial derivatives of the function at the specified points.

$$f(x, y) = \sqrt{2x + 3y - 1}, \ \frac{\partial f}{\partial x} \text{ and } \frac{\partial f}{\partial y} \text{ at } (-2, 3).$$

Problem 4. Let $w = f(x, y, z)$ be a function of three independent variables. Write down a definition of the partial derivative $\partial f/\partial y$ at (x_0, y_0, z_0) and use it to find $\partial f/\partial y$ at $(-1, 0, 3)$ for $f(x, y, z = -2xy^2 + yz^2$.

Problem 5. (a) First express dw/dt as a function of t, both by using the Chain Rule and by expressing w in terms of t and differentiating directly with respect to t.

(b) Then evaluate dw/dt at the given value of t.

$$w = 2ye^x - \ln z, \quad x = \ln(t^2 + 1), \quad y = \tan^{-1} t, \quad z = e^t; \quad t = 1.$$

In Problems 6–7, find partial derivatives at specified points.

Problem 6. Find $\partial w/\partial r$ when $r = 1$, $s = -1$ if $w = (x + y + z)^2$, $x = r - s$, $y = \cos(r + s)$, $z = \sin(r + s)$.

Problem 7. Find $\partial z/\partial u$ and $\partial z/\partial v$ when $u = \ln 2$, $v = 1$ if $z = 5 \tan^{-1} x$ and $x = e^u + \ln v$.

Problem 8. The pressure P (in kilopascals), volume V (in liters), and temperature T (in kelvins) of a mole of an ideal gas are related by the equation

$$PV = 8.31\,T.$$

Find the rate at which the pressure is changing when the temperature is 300 K and increasing at a rate of 0.1 K/s and the volume is 100 L and increasing at a rate of 0.2 L/s.

Problem 9. Assuming that the equation below defines y as a differentiable function of x, find the value of dy/dx at the given point.

$$x^2 + xy + y^2 - 7 = 0, \quad (1, 2).$$

Problem 10. Assuming that the equation below defines z as a differentiable function of x and y, find the value of $\partial z/\partial x$ and $\partial z/\partial y$.

$$x^3 + y^3 + z^3 + 6xyz = 1.$$

END

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Answers to selected problems

1. 1; $\quad -\dfrac{y}{\sqrt{y^2+z^2}}; \quad -\dfrac{z}{\sqrt{y^2+z^2}}$
3. 1/2; $\quad 3/4$
4. 9
5. (b) $1 + \pi$
6. 12
8. $\approx$ 0.042 kPa/s
9. 4/5